Assessment of a Selection–Plan 3 Hybrid Graph Entry

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1 The Proposed Replacement

The proposed route is:

selection $\Rightarrow$ controlled class $\Rightarrow$ Plan 3 good inset level $E_{\hat{t}} \Rightarrow L^2$-Serrin $\Rightarrow L^\infty$-Serrin $\Rightarrow$ Brandolini annulus $\Rightarrow L^\infty$

The conclusion of this check is mixed. The first half is genuinely useful: selection turns the weighted D_H variance into honest unweighted L^2 Serrin control on the stopped level, and the inward cut preserves the bad quotient/asymmetry. The weak point is the $L^2 \to L^\infty$ upgrade. It is not a consequence of Lipschitz boundary geometry alone.

2 What Selection Gives

Let U be the selected minimizer from Plan 1, with torsion function u. The BDV/selection regularity package gives a uniform Lipschitz bound for the potential:

$$\|\nabla u\|_{L^\infty(U)} \le L_*(N,R). \tag{2.1}$$

It also gives density and nondegeneracy. Before the flatness/graph-entry step, it does *not* give a global Lipschitz parametrization of ∂U. Thus the statement that selection reduces to Lipschitz boundary is stronger than the current Plan 1 selection lemma. If one assumes a uniform Lipschitz boundary class separately, the discussion below is conditional on that assumption.

3 Chebyshev Level Gives Honest L^2

Apply the Plan 3 volume-variable Chebyshev selection inside U. For every fixed small $\theta > 0$, there is a regular level

$$E = E_{\hat{t}} := \{u > \hat{t}\}$$

such that

$$|U \setminus E| \le C_\theta \delta_T(U), \qquad D_H(\hat{t}) \le C\theta. \tag{3.1}$$

On this level the exact identity gives, with $f = |\nabla u|$ and $\bar{f} = |E|/P(E)$,

$$\int_{\partial E} \frac{(f-\bar{f})^2}{f}\, d\mathcal{H}^{N-1} = \frac{|E|}{P(E)^2} D_H(\hat{t}). \tag{3.2}$$

Using (2.1),

$$\begin{aligned}\int_{\partial E}(f-\bar f)^2\,d\mathcal{H}^{N-1} &= \int_{\partial E} f\,\frac{(f-\bar f)^2}{f}\,d\mathcal{H}^{N-1}\\ &\le L_*(N,R)\int_{\partial E}\frac{(f-\bar f)^2}{f}\,d\mathcal{H}^{N-1}\le C(N,R)\theta.\end{aligned}\tag{3.3}$$

This is the positive part of the hybrid route: selection removes the upper-gradient tail obstruction that raw D_H had.

4 The Inward Cut Preserves Asymmetry

Let

$$q := \frac{E(U)-E(B)}{\alpha(U)}.$$

Since $\delta_T(U) = q\,\alpha(U)$, (3.1) gives

$$|U\setminus E|\le C_\theta q\,\alpha(U).$$

The smooth asymmetry α is L^1-Lipschitz on bounded sets. After volume normalization of E,

$$\alpha(E^\sharp)\ge \alpha(U)-C|U\setminus E|-C|\,|E|-|U|\,|\ge(1-C_\theta q)\alpha(U).\tag{4.1}$$

Thus in the selection contradiction regime $q\ll\theta$, the inset level keeps the asymmetry comparable to that of U. This part of the proposed route is sound.

5 The Missing $L^2\to L^\infty$ Upgrade

To use a Brandolini/Serrin annulus theorem one wants

$$\|\,|\nabla u|-\bar f\,\|_{L^\infty(\partial E)}\le\varepsilon_B\tag{5.1}$$

with $\varepsilon_B\ll 1$. The L^2 estimate (3.3) implies this only if one has a uniform modulus of continuity for $f=|\nabla u|$ on ∂E. For instance, if

$$[f]_{C^\gamma(\partial E)}\le M\tag{5.2}$$

and ∂E has uniformly controlled $(N-1)$-dimensional geometry, then interpolation gives

$$\|f-\bar f\|_{L^\infty(\partial E)}\le C\,M^{\frac{N-1}{N-1+2\gamma}}\,\|f-\bar f\|_{L^2(\partial E)}^{\frac{2\gamma}{N-1+2\gamma}}.\tag{5.3}$$

Then choosing θ small enough would give (5.1).

The problem is that (5.2) is not implied by a uniform Lipschitz boundary. Lipschitz domains do not give uniform continuity of ∇u up to the boundary. For the inset level ∂E, qualitative analyticity is free because the level is inside U, but the quantitative constants depend on

$$\operatorname{dist}(\partial E,\partial U)\quad\text{and}\quad\inf_{\partial E}|\nabla u|.$$

The stopped level is only $O_\theta(\delta_T)$ from the boundary in measure, and the physical distance can go to zero. Also D_H controls the low-gradient set only in measure:

$$\mathcal{H}^{N-1}\{|\nabla u|\le s\}\lesssim D_H\,s,$$

not pointwise. Hence it does not give a uniform lower bound for $|\nabla u|$ on the entire level.

Therefore the implication

$$\text{selected Lipschitz class} + D_H(\hat{t}) \ll 1 \Longrightarrow \|\, |\nabla u| - \bar{f}\,\|_{L^\infty(\partial E)} \ll 1$$

is currently unproved and is false as a general Lipschitz-domain principle. It becomes true if one adds a uniform C^γ modulus for the boundary gradient on ∂E, but that is essentially a free-boundary regularity input.

6 Inset Boundary Regularity

For a regular value $\hat{t}$, $\partial E = \{u = \hat{t}\}$ is a smooth embedded hypersurface. This is qualitative. A quantitative Lipschitz constant follows from the implicit function theorem with constants depending on

$$\frac{\|D^2 u\|_{L^\infty(\text{collar})}}{\inf_{\partial E} |\nabla u|}.$$

Neither term is uniformly controlled by (2.1) and $D_H(\hat{t}) \ll 1$. Thus "the inset boundary is Lipschitz" is true with constants depending on the selected level, but not with the uniform constants needed for a proof.

7 Brandolini Annulus and Graph Entry

Assume conditionally that (5.1) is obtained and that ∂E lies in the geometric class required by a Brandolini-type stability theorem. Then one gets a thin annulus:

$$B_{r_i}(z) \subset E \subset B_{r_e}(z), \qquad r_e - r_i \le \eta_B. \tag{7.1}$$

This gives Hausdorff closeness to a ball. It does not by itself give a radial graph. A set may contain $B_{r_i}(z)$, be contained in $B_{r_e}(z)$, and still have overhangs or extra components in the thin shell. One still needs a one-crossing/star-shapedness theorem, or an $O(\eta_B)$-cost radialization.

8 Conditional Version That Would Work

The proposed route becomes correct under the following additional hypotheses:

1. the selected/inset level belongs to a uniform geometric class for which $|\nabla u|$ has a C^γ modulus bounded by $M(N, R)$;

2. Brandolini's annulus theorem applies to that class;

3. thin annulus plus selected topology implies one radial crossing, or an annulus-to-radialization theorem is available.

Under these assumptions, choose θ below the interpolation and Brandolini thresholds. Chebyshev gives (3.3); interpolation gives (5.1); Brandolini gives the annulus; the final one-crossing/radialization input gives a small L^∞ radial graph. Equation (4.1) and the superlevel torsion transfer preserve the bad quotient.

9 Verdict

The hybrid has a useful core:

$$\text{selection's Lipschitz bound for } u + D_H \text{ on a Chebyshev level} \Longrightarrow L^2\text{-Serrin on that level.}$$

It also preserves asymmetry in the selection contradiction regime.

But it does not yet replace the Alt–Caffarelli graph-entry machinery. The serious missing theorem is a uniform $L^2 \to L^\infty$ Serrin upgrade on the inset level. Merely assuming Lipschitz boundary is not enough; one needs a uniform modulus of continuity for the boundary gradient, which is effectively regularity/flatness information. A second, smaller missing theorem is the annulus-to-radial-graph or annulus-to-radialization step.

Thus the best current formulation is not a complete replacement for Plan 1, but a narrowed target: prove just enough regularity of the *inset level* to make the interpolation estimate (5.3) valid, instead of proving global graph regularity of ∂U directly.